

SMART DISPLAY MODULE SPECIFICATION

3.5 Inch Smart Display with TOUCH	
Model:	UEED32480035E-WB-A
Version:	V1.0
Date:	2026-08-08

Customer Confirmation

Approved by	Notes

REVISION HISTORY

Revision	Date	Contents of Revision Change	Remark
V1.0	20260808	Preliminary release	

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1. Introduction

3.5-inch Transflective 480*320 ESP32 Smart Display



Front - Capacitive Touch Display



Back - ESP32 Dev Board

Brand	VIEWE	Model	PLCM-UEED32480035E-WB-A
Size	3.5inch 480*320	Processor	ESP32-S3 240Mhz
Viewing Angle	IPS TFT All View	PSRAM Memory	8MB
Brightness	112+12% Ambient Light cd/m2	Flash Memory	16MB
Interface	MCU 8/16BIT/SPI 40pin	Wireless Module	2.4GHz Wi-Fi, BLE 5.0, BLE Mesh
Touch Type	Capacitive Touch Panel CTP	Expansion	SD Card, TYPE-C, IO, UART, RS485

UEED32480035E-WB-A is a high-performance smart display development board based on ESP32-S3. It is equipped with a 3.5-inch SPI transflective touch screen (320×480 resolution), whose screen remains clearly visible under sunlight, making it an ideal choice for outdoor devices. Designed by VIEWE, this development board is suitable for IoT and outdoor Human-Machine Interface (HMI) application scenarios that require abundant peripheral interfaces as well as Wi-Fi/Bluetooth Low Energy connectivity.

1.1 Product Features

CPU:

1. Processor

- 1) Equipped with an Xtensa 32-bit LX7 dual-core processor, with a main frequency up to 240 MHz.
- 2) Integrated Wi-Fi 2.4GHz (802.11 b/g/n) and Bluetooth 5 (LE) & BLE Mesh.

2. Memory

- 1) 8 MB PSRAM.
- 2) 16 MB Flash.
3. Peripheral Interfaces
 - 3) Two 2*20 Pin Headers are on-board, breaking out multiple programmable GPIOs, supporting SPI, UART, I2C, I2S, LCD, Camera, USB OTG, and other interfaces.
 - 4) On-board USB Type-C port for power supply, programming, and serial debugging (CH340C).
 - 5) On-board Micro SD card slot (SPI interface).
 - 6) Reset and Boot buttons.

For more information on ESP32-S3-WROOM-1, please refer to the following link:[datasheet_en.pdf](#)

Display

- 1) Size: 3.5 Inch
- 2) Resolution: 320(W)*3(GB)*480(H)
- 3) Mode: MCU 8/16BIT/SPI
- 4) Display mode: Transflective Normally BLACK
- 5) Driver IC: ST7365 or Compatible
- 6) Touch IC: CH340
- 7) Backlight Type: White LED

Other

- 1) Operation Temperature: -20~70°C
- 2) Storage Temperature: -30~80°C

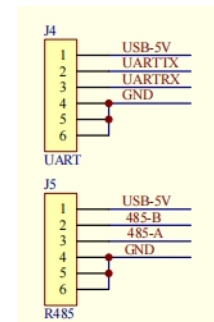
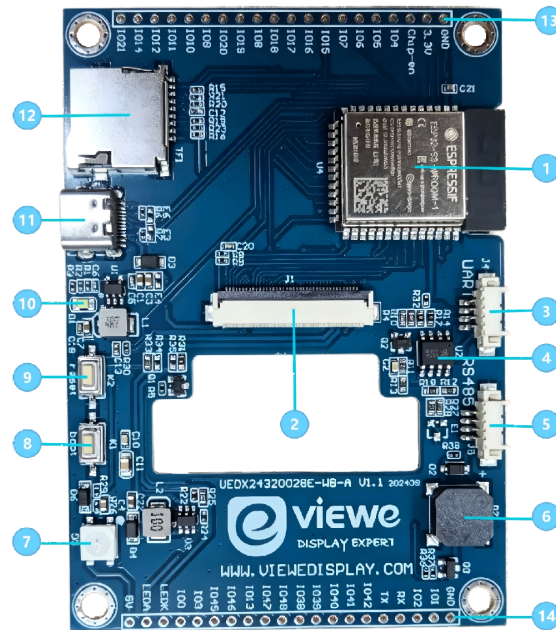
1.2 Applications

With rich connectivity, powerful processing capabilities and outdoor-oriented reliability, the UEDX32480035E-WB-A is an ideal choice for outdoor-deployed IoT devices in the following areas:

- Outdoor Smart Control Panels
- Field Industrial Automation HMI
- Outdoor Smart Appliances
- Outdoor-rated Consumer Electronics
- Field Wireless Data Loggers
- Outdoor Touch Screen Interfaces
- Outdoor Educational Learning Platforms

2. Product information

2.1 Interface Description



1、Main Control Chip: ESP32S3-MCN16R8

Dual-core processor, up to 240MHz operating frequency

2、Display Interface:

SPI interface adapted for 3.5- inch semi- transparent reflective screen, plug- and- play. Please refer to the following Display Interface table for specific I/O details and descriptions.

3、UART Serial port

It is recommended to use the matching HY1.25 4P- to - Dupont male header. But the RS485 chip(MS1285) numbered 4 needs to be removed during use. This is the current bug of this board.

4、MS1285

MS1285 is transceiver chip applied to RS-485 communication systems.

5、RS485 Serial port

It is recommended to use the matching HY1.25 4P- to - Dupont male header.

6、Active Buzzer

7、RGB-LED (WS2812B)

8、Boot Buttons

BOOT button (GPIO0) for firmware download mode.

9、Reset Buttons

RESET button (CHIP-EN) for resetting the board.

10、Power indicator light

11、USB Type-C interface

Used for 5V DC power supply

12、SD Card Slot

SPI interface connected for external storage expansion

13 14、External GPIO Headers

Dual-row 2*21 headers providing access to a wide range of GPIOs, including ADC, touch sensors, and standard digital I/O

Currently, this board has certain vulnerabilities. The UART receive pin is overwritten by RS485. If UART must be used, there are two solutions:

- 1、 Use the extended pins for UART
- 2、 Remove the RS485 transceiver chip

Display Interface

Pin No.	Symbol	I/O	Description
1	XL/CTP-SCL	I	I2C clock signals for CTP; Option XL for RTP
2	YU/CTP-SDA	I	I2C data signal for CTP,Option YU for RTP
3	XR/CTP-RST	I	The signal will reset the CTP,Signal is active low, Option XR for RTP
4	YD/CTP-INT	I/O	Interrupt signals for CTP,Option YD for RTP
5	GND	P	Power Ground
6	IOVCC	P	Power supply for I/O system
7	VCI	P	Power supply for analog circuits
8	TE	O	Tearing effect signal is used to synchronize MCU to frame memory
9	SPI_CS/MCU_CS	I	Chip selection pin. Low-active
10	SPI_SCL/MCU_RS	I	Display data/command selection pin in MCU interface In SPI mode, this pin is used as SCL
11	SPI_RS/MCU_WR	I	Write enable in MCU parallel interface RS=1 display data or parameter;RS=0 register index / command
12	MCU_RD	I	Read enable in 8080 MCU parallel interface. Low-active.
13	SPI_SDA	I/O	Serial communication data input and output, internal pull low.
14	SPI_SDO	O	SPI interface output pin
15	RESET	I	The signal will reset the LCM, Signal is active low.
16	GND	P	Power Ground
17-32	DB0-DB15	I/O	data bus for MCU
33	LED-A	P	Power supply for backlight anode
34-36	LED-K	P	Power supply for backlight cathode
37	GND	P	Power Ground
38	IM0	I	The MCU interface mode select.
39	IM1	I	The MCU interface mode select.
40	IM2	I	The MCU interface mode select.

The choice of IM is as follows: 1 means pull up, and 0 means pull down

IM2IM1IM0: 110-2.4inch 111-3.5inch 101-3.9/4.3inch

2.2 GPIO Definition

Symbol	Current Usage
CHIP-EN	RESET
GPIO0	BOOT
GPIO0	RGB-LED
GPIO1	TP-SDA
GPIO2	TP-RST
GPIO3	TP-SCL
GPIO4	TP-INT
GPIO5	GPIO5
GPIO6	GPIO6
GPIO7	GPIO7
GPIO8	GPIO8
GPIO9	GPIO9
GPIO10	GPIO10
GPIO11	GPIO11
GPIO12	GPIO12
GPIO13	BL-EN
GPIO14	SD-SCLK
GPIO15	SD-D2
GPIO16	SD-MISO
GPIO17	SD-MOSI
GPIO18	SD-D1
GPIO19	USB-DN
GPIO20	USB-DP
GPIO21	SD-D3
GPIO38	GPIO38
GPIO39	LCD-RST
GPIO40	LCD-SCL
GPIO41	LCD-RS
GPIO42	LCD-CSB
GPIO43	UARTTX/RS485
GPIO44	UARTRX/RS485
GPIO45	LCD-SDA
GPIO46	LCD-SDO
GPIO47	LCD-M0
GPIO48	LCD-M1

Button

LED

GPIO

Touch

LCD

UART/RS485

SD

USB

2.3 Display Information

Item	Specification	Unit	Remark
Pixel Driving element	TFT	-	-
Screen Size	3.47	Inch	Diagonal
Resolution	320(H)*3(RGB)*480(V)	Dots	-
Interface	MCU 8/16BIT/SPI	-	40PIN
Module Power Consumption	0.07	Watt	Reflective Mode
	0.42		Transmissive Mode
Luminance	12%*Surrounding illumination	cd/m ²	Reflective Mode
	150+12%*Surrounding illumination	cd/m ²	Transmissive Mode
Active Area	48.96(W)*73.44(H)	mm	-
Pixel pitch (W*H)	0.153(W)*0.153(H)	mm	-
Module Size (W*H*D)	52.26(W)*80.6(H)*2.21(D)	mm	Typ

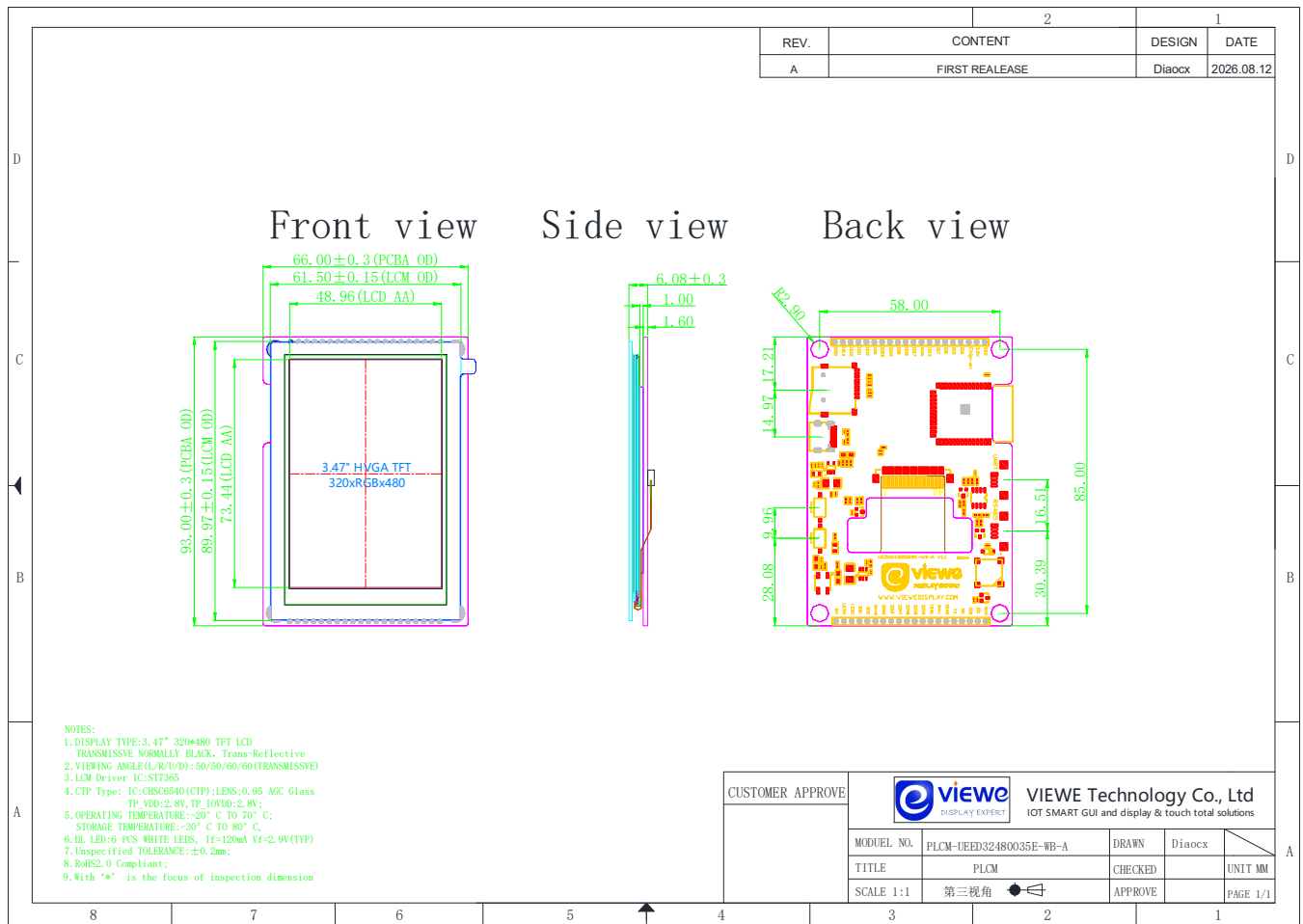
2.4 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	4.0	5.0	5.5	V
Operation Current	VCC= +5V, Maximum backlight current	-	200	-	mA
	VCC= +5V,backlight off	-	100	-	mA
Recommended power supply:5V 1A DC					

2.5 Reliability Test

Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	70	C
Storage Temperature	---	-30	25	80	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

3. MECHANICAL DRAWING



4. Related downloads

GitHub: [VIEWESMART/UEED32480035ESP32-3.5inch-320 480-Transflective-Display](https://github.com/VIEWESMART/UEED32480035ESP32-3.5inch-320-480-Transflective-Display)

Gitee: [VIEWESMART/UEED32480035ESP32-3.5inch-320 480-Transflective-Display](https://gitee.com/VIEWESMART/UEED32480035ESP32-3.5inch-320-480-Transflective-Display)